RUBIERE **RAMIREZ**

Geospatial Specialist

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Belmopan City, Cayo, Belize

EMPLOYMENT HISTORY

Geographic Information Systems Specialist Consultant,
Coral Research and Development Accelerator Platform,
Fragments of Hope, Belize

December 2025 – Present
Placencia, Belize

- Design and implement remote sensing projects using high-resolution satellite and drone derived imagery.
- Conduct thorough assessments of FoH needs and provide expert recommendations for GIS solutions and best geodata management practices.
- Create and maintain detailed documentation, such as project plans, technical specifications, and user manuals, to ensure seamless project execution.
- Provide training and support to FoH on geospatial technologies and methods in fostering their ability to utilize spatial data effectively.

Geographic Information Systems Assistant,
Systems Development and Data Processing,
Statistical Institute of Belize.

September 2022 – Present
Belmopan City, Belize

- Supervise the GIS Technician and Data Processing Technicians.
- Create, administer and maintain spatial databases on PostgreSQL database with PostGIS extension.
  - Utilize SQL and Spatial SQL within PostgreSQL with PostGIS to automate workflows by proficiently creating databases, schemas, tables, views, procedures, functions, triggers, events, users, roles, and efficiently granting privileges, ensuring streamlined access management and comprehensive data administration.
- Create, administer and maintain web applications on free and open-source software for geospatial (FOSS4G): GeoServer/Mapstore2.
- Deploy, administer and maintain web applications on commercial off-the-self (COTS) software: Portal for ArcGIS and ArcGIS Server; and Oracle Application Express.
- Perform earth observation on the built environment: create Sentinel-2 geomedian satellite imagery with Google Earth Engine and use machine learning algorithms to predict built up areas.
- Create tools to automate and standardize the extraction, transformation, and loading (ETL) of non-spatial data and spatial data into spatial data repositories with SQL, R and Python.
- Create spatial features with R programming language and Python with Open Geospatial Consortium (OGC) Standards.
- Research and develop systems design for geospatial solutions at the Institute.
- Provide training and support to the Geospatial Information Unit and other departments at the Institute.
- Created a full stack PostgreSQL, Express, React, NodeJS web application to geovisualize spatial data.
- Created *{rvr2spatial}* a package in R programming designed to facilitate the ETL process from MariaDB to PostgreSQL database with PostGIS extension for the Institute's needs. The package also does quality assurance and quality control on the Institute's spatial data. The package can be inspected here: https://github.com/Rubi3r3/rvr2spatial.

- Created *MariaDB to QGIS* which is a QGIS Python Plugin to fetch longitude (x) and latitude (y) data from MariaDB and writes the table to a shapefile or geopackage. The plugin also handles data without geometry, saving it to a specified directory as a comma separated value (CSV) or as a table within the geopackage. Additionally, the plugin also adds the points to the QGIS map canvas and displays data without geometry as CSV or table into the layers pane. The MariaDB to QGIS plugin is published on the QGIS Python Plugin Repository: https://plugins.qgis.org/plugins/qgis_mariadb_plugin, and the source code can be inspected here: https://github.com/Rubi3r3/qgis_mariadb_plugin.

Geographic Information Systems Technician, July 2018 – September 2022
Systems Development and Data Processing, Belmopan City, Belize
Statistical Institute of Belize.

- Heads-up digitizes building footprints in ArcGIS suite of products and QGIS from Google Earth Pro and Environmental Systems Research Institute's World Imagery, Bing Maps, Open Street Maps, Satellite pour l'Observation de la Terre 7 (SPOT-7).
- Provide both digital web maps and hard copy maps for different departments at the Statistical Institute upon request.
- Perform remote sensing techniques to detect changes in urban areas in QGIS.
- Process aerial photos from drone imagery in QGIS and ArcGIS Pro.
- Assist with the preparation of Enumeration District Maps for the 2022 Population and Housing Census.
- Assist with the creation of maps in the SIB Census Dashboard with R programming and RStudio.
- Perform quality assurance and quality control on census spatial field data collected using ArcGIS, QGIS and R Programming.
- Automate the generation of spatial data with R programming language using OGC Standards.
- Implement Row level and column level security with table views on PostgreSQL database with PostGIS Extension.
- Implement Geofence, Geofence Data Rules and Geofence Admin Rules on Open-Source GeoServer.
- Implement Row level and column level security with table views on PostgreSQL database with PostGIS Extension.

Geographic Information Systems Technician, May 2017 – June 2017
Coastal and Marine Data Centre, Belize City, Belize
Coastal Zone Management Authority and Institute of Belize

- Updated building footprints for Caye Caulker using Google Earth Pro and existing building footprints data.
- Prepare maps for ground-truthing in Caye Caulker using updated building footprints and parcel data provided by the Land Information Center, Ministry of Natural Resources.
- Assisted with the collection of land use and building information in Caye Caulker using Open Data Kit for android devices.
- Processed data obtained from the open data kit for Caye Caulker using the building footprints Urban Land Use Data Processing Manual standardized by the Land Information Center at the Ministry of Natural Resources.

Stann Creek Field Representative, 2015 – 2016
Oceana in Belize Stann Creek, Belize

- Represented Oceana Belize in the Stann Creek district with the public, community-based organizations, non-governmental organizations, and the business sector.

- Conducted presentations about Oceana in Belize at the primary, secondary and tertiary level schools as well as to the community and business sector in the district.
- Canvassed neighborhoods in the district and collected people's sentiments on offshore oil exploration and exploitation in Belize as well as collect signatures against offshore oil exploration and exploitation.
- Recruited wavemakers (volunteers) in Stann Creek district.
- Maintain and update a wavemaker database for the Stann Creek district.
- Organized public events in the district such as beach cleanups, mural unveiling, people's referendum anniversary event, among others.
- Procured venders for community projects, provided logistics and reached out to the community for public events.

EDUCATION

MSc in Geographical Information Science and Systems *(In Progress)*
Department of Geoinformatics - Z_GIS,
Faculty of Digital and Analytical Sciences,
Paris Lodron University of Salzburg

2026
Salzburg City, Austria

Completed coursework includes:

- Introduction to Geoinformatics
- Data Modeling and Data Structures
- Data Sources and Data Acquisition
- Geodatabase Management
- Basics of Application Development
- Remote Sensing: Basic
- Remote Sensing: Advanced
- Geographical Analysis
- Automated Data Processing with R
- Spatial Statistics

BSc in Natural Resource Management
Department of Science,
Faculty of Science and Technology,
University of Belize

2014
Belmopan City, Belize

Associate in Natural Resources Management
Department of Science,
Faculty of Science and Technology,
University of Belize

2011
Belmopan City, Belize

Associate in Arts
St. John's College Junior College
Majors: General Studies and Chemistry

2009
Belize City, Belize

High School Diploma
St. John's College High School
Concentration: Biology, Chemistry and Physics

2007
Belize City, Belize

SKILLS

- R
- R-Spatial
- Python
- GeoPython
- SQL
- Spatial SQL
- QGIS
- ArcGIS Pro
- Spatial Data Science
- Data Engineering
- Remote Sensing & Earth Observation
- Google Earth Engine
- Oracle Spatial
- PostgreSQL with PostGIS extension
- FOSS4G GeoServer
- Geovisualization & Geocommuniation
- Web GIS / Internet GIS

ACHIEVEMENTS

✓ **Korea International Cooperation Agency (KOICA) and the Statistical Institute of Belize Scholarship**
Awarded under the *Enabling Belize's Statistical System Project* to pursue a postgraduate degree in Geographical Information Science and Systems.

✓ **Geospatial Database Management System: ArcSDE, QGIS and PostgreSQL PostGIS Integration**
Developed and implemented an ArcSDE/Multiuser PostgreSQL database with PostGIS extension for environment agnostic editing in ArcGIS Pro and QGIS.

✓ **Geospatial Database Management System: Oracle database with locator/spatial extension (Oracle Spatial)**
Developed and implemented an instance of Oracle Spatial to serve spatial data into Oracle Application Express with Oracle's SDO_Geometry from PostgreSQL database with PostGIS extension using GDAL – ogr2ogr and reconciled with python programming.

✓ **Enterprise GIS in FOSS4G**
Developed and implemented Enterprise Geographic Information Systems infrastructure exclusively with free and open-source software for geospatial: QGIS and PostgreSQL Database with PostGIS extension.

✓ **Web GIS in FOSS4G**
Deployed and maintained Open-Source GeoServer and Mapstore2 with Apache Tomcat on Apache HTTP Server on SIB's GIS Servers.

✓ **Web GIS Development: FOSS4G Web Application for Census Progress Tracking**
Geovisualized census progress coverage by creating a web application using open standards: Mapstore2 Web Application created using GeoServer, PostgreSQL database with PostGIS extension and ETL process automated with R programming.

✓ **Web GIS Development: Oracle Application Express**
Developed and managed a GeoPortal that uses a Python API to access PostgreSQL geospatial data within Oracle Application Express.

✓ **Web GIS Development: Full-Stack PERN Application**

Built and maintained a PostgreSQL with PostGIS extension, Express.js, React, Node.js geoportal to geovisualize the Statistical Institute of Belize's spatial data.

✓ **Created {rvr2spatial}**
A package in R programming that was designed to facilitate the ETL process from SIB's MariaDB to PostgreSQL database with PostGIS extension.

PROFESSIONAL DEVELOPMENT

- **R for Official Statistics: A Crash Course,** Facilitated by the United Nations Statistical Institute for Asia and the Pacific, 3rd June to 5th June 2025.

- **Geospatial Capacity Building for Censuses and Surveys with Open-Source Tools - QGIS, R and Survey Solutions,** Facilitated by the United Nations Population Fund (UNFPA) and the Korea International Cooperation Agency (KOICA), 19th to 23rd May 2025.

- **Principles of Reproducible Analytical Pipelines for Official Statistics,** Facilitated by United Nations Statistical Institute for Asia and the Pacific (SIAP), 17th February to 14th March 2025.

- **Urban Area Classification through Remote Sensing and Machine Learning Techniques,** Facilitated by the Mexican National Institute of Statistics and Geography (INEGI), 27th to 24th May 2024.

- **Food Insecurity Experience Scale (FIES) and Sustainable Development Goal 2.1.2 Analysis Workshop,** Facilitated by the Food and Agriculture Organization (FAO) of the United Nations Statistics Division, 6th May to 10th May 2024.

- **A National Validation Workshop on Advancing Comprehensive Disaster Management (CDM) Blueprint Implementation**, Facilitated by the Caribbean Disaster Emergency Management Agency (CDEMA), Natural Disaster Facility (NDF) in the CARIFORM Programme, the European Union (EU), & the National Emergency Management Organization (NEMO), 11th October 2023.

- **III Workshop on the use of Big Data for Official Statistics,** Facilitated by the United Nations Big Data Regional Hub in Brazil, United Nations Statistics Division, Brazilian Institute of Geography and Statistics (IBGE), National Administrative Department of Statistics (DANE) Colombia, National Institute of Statistics (INE) Paraguay, 31st July, 2nd August and 4th August 2023

- **Caribbean Geospatial Webinar – End User Training,** Facilitated by the Caribbean Community Climate Change Centre and Caribbean Development Bank, September 2021.

- **Advance Spatial Data Analysis Using ArcGIS,** Facilitated by the Belize National Spatial Data Infrastructure, Ministry of Natural Resources and Total Business Solutions Ltd, October 2018.

- **Creative Communications Training,** Facilitated by ArtCorps, May 2015.

- **Leadership Training for Social and Environmental Justice,** Facilitated by the Belize Institute of Environmental Law and Policy, August 2014.

References

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Science Department,
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